

KADI SARVA VISHWAVIDYALAYA
B.E 7th SEMESTER EXAMINATION (November 2025)
SUBJECT: Blockchain Technology (Code: CT703D-N)

Date:13/11/2025

Time: 12:30 PM to 03:30 PM

Marks: 70

Instruction:

1. Answer each section in separate Answer Sheet.
2. All questions are compulsory.
3. Indicate **clearly**, the option you attempted along with its respective question number.
4. Use the last page of main supplementary for rough work.

Section:1

- Q.1** (a) What is a cryptographic hash function, and how does SHA fit into this category? [05]
(b) How does cryptographic security differ from traditional security measures? [05]
(c) What is a Merkle tree, and how does it organize data within a block? [05]

OR

- (c) What are the key differences between centralized and distributed networks? [05]
Q.2 (a) What is Bitcoin, and how does it function as a decentralized digital currency? [05]
(b) How does each field in the block structure contribute to blockchain functionality? [05]

OR

- Q.2** (a) What is a double-spending attack in the context of Bitcoin? [05]
(b) What is a distributed ledger, and how does it differ from traditional ledgers? [05]

- Q.3** (a) What are the main differences between permissioned and permission-less blockchain models? [05]
(b) What is an orphan block, and how does it occur in a blockchain network? [05]

OR

- Q.3** (a) What is Proof of Work (PoW), and how does it relate to mining? [05]
(b) How do Denial of Service (DoS) attacks threaten blockchain nodes or networks? [05]

Section:2

- Q.4 (a) Explain the role and responsibilities of a miner. [05]
(b) Explain the advantages and disadvantages of mining pools. [05]
(c) Explain real-world use cases of permissioned blockchains. [05]

OR

- (c) Explain the scalability issues faced by permissioned blockchains. [05]
Q.5 (a) What is the Three Phase Commit Protocol? Why was it introduced? [05]
(b) Explain the difference between solo mining and pool mining. [05]

OR

- Q.5 (a) What is blockchain-based crowdfunding? Explain in detail. [05]
(b) Discuss how blockchain reduces fraud and counterfeit goods in supply chains. [05]
Q.6 (a) How do smart contracts differ from traditional legal contracts? [05]
(b) What is the role of nodes, miners, and consensus in Ethereum's architecture? [05]

OR

- Q.6 (a) What is gas in Ethereum? Why is it required? [05]
(b) Discuss data types and functions in Solidity. [05]

KADI SARVA VISHWAVIDHYALAYA
B.E. Semester VII Examination – March – 2025
CT703D-N Blockchain Technology

DATE: 01/04/2025

TIME: 12:00 pm to 03:00 pm

TOTAL MARKS: 70

Instructions:

- 1 Answer each section in separate answer sheet.
- 2 Use of scientific calculator is permitted.
- 3 All questions are **Compulsory**.
- 4 Indicate **clearly**, the options you attempt along with its respective question number.
- 5 Use the last page of main supplementary for **rough work**.

SECTION – I

- Q-1 A Explain Types of Blockchain. 5
B Differences among centralized, decentralized and distributed system. 5
C What is Merkle root. Explain concept of Merkle tree. 5

OR

- C Write short note on public key cryptography. 5

- Q-2 A Write short note in Bitcoin. 5
B Explain double spending attack. 5

OR

- A Write short note in Bitcoin scripts. 5
B Explain the concept of orphan block. 5

- Q-3 A Explain the features of cryptographic hash functions. 5
B Explain Genesis block. Explain typical block structure. 5

OR

- A Write short note on proof of stake. 5
B Explain block propagation. 5

SECTION – II

- Q-4 A Explain consensus in Blockchain technology. 5
B Write short note on proof of work. 5
C Explain sybil and DDOS attack. 5

OR

- C What is a mining pool? 5

- Q-5 A What is mining. Explain the term mining difficulty. 5
B Write short note on RAFT consensus algorithm. 5
OR
A Write short note on PAXOS consensus algorithm. 5
B Explain Three Phase Commit Protocol. 5
- Q-6A Explain Practical Byzantine Fault Tolerant model. 5
B Explain the role of Blockchain in supply chain management. 5
OR
A Write short note on smart contract. 5
B Explain the role of Blockchain in crowd funding. 5

KADI SARVA VISHWAVIDYALAYA

BE. Semester VII

Blockchain Technology - CT703D-N

Date: 19/11/2024

Time: 12:30 to 3:30

TOTAL MARKS: 70

Instructions:

- 1 Answer each section in separate answer sheet.
- 2 Use of scientific calculator is permitted.
- 3 All questions are **Compulsory**.
- 4 Indicate **clearly**, the options you attempt along with its respective question number.
- 5 Use the last page of main supplementary for **rough work**.

SECTION – I

Q-1 Answer the following questions.

- A What is Blockchain? Explain Genesis Block in Blockchain network. 5
- B Explain core components of blockchain architecture. 5
- C Explain which one is better from the decentralized and distributed system. 5

OR

- C What is orphan block in Blockchain network? 5

Q-2 Answer the following questions.

- A What is consensus mechanism in blockchain network? List out consensus mechanism algorithm. 5
- B Explain POW, POS and Proof of Burn. 5

OR

- A What is Bitcoin and How does it work? 5
- B What is the difference between distributed Ledger and Blockchain. 5

Q-3 Answer the following questions.

- A Explain use of digital signature and public key cryptography in blockchain system. 5
- B Explain cryptographic hash function. 5

OR

- A Explain Sybil attack and DDOS attack. 5
- B Write short note on permissioned and permissionless blockchain. 5

SECTION – II

Q-4 Answer the following questions.

- A Explain the concept of smart contracts and give one real-world use case 5
- B What is the Byzantine Generals Problem and why is it significant in blockchain? 5
- C Explain how blockchain can be used in crowdfunding and its potential benefits. 5

OR

- C Compare the PAXOS and RAFT consensus algorithms in detail. 5

Q-5 Answer the following questions.

- A What role does blockchain play in supply chain management? 5
- B What is a Merkle Tree and how does it enhance data integrity within a blockchain? 5

OR

- A Discuss the advantages of using blockchain technology in international trade finance. 5
- B What is Bitcoin Script (FORTH) and how does it enable Bitcoin transactions? 5

Q-6 Answer the following questions.

- A Explain the concept of a smart contract and its significance in Ethereum. 5
- B What is Solidity and why is it used for developing Ethereum smart contracts? 5

OR

- A What is "gas" in Ethereum and why is it important? 5
- B Describe the role of the Ethereum Virtual Machine (EVM) in executing smart contracts. 5

Exam Number: _____

KADI SARVA VISHWAVIDYALAYA
B.E. Semester VII (REG/ ATKT) EXAMINATION October 2024

Subject Name : Blockchain Technology

Subject Code: CT703D-N

Date: 21/10/2024(Monday)

Time: 12.30 pm to 03.30 pm

Total marks: 70

Instructions:

1. Answer each section in separate Answer sheet.
2. All questions are compulsory.
3. Indicate clearly, the options you attempt along with its respective question number.
4. Use the last page of main supplementary for rough work.

Section-I

Q.1(A) What are the different types of Blockchain? (5)

Q.1(B) Explain the concept of centralization, decentralization and distributed. (5)

Q.1(C) What are the Merkle trees? What is its importance in Blockchain? (5)

OR

Q.1(C) What are the Characteristics of cryptographic hash function. (5)

Q.2 (A) What is bitcoin? How did it evolve? (5)

Q.2(B) Explain the double spending attack. (5)

OR

Q.2 (A) What are the features of bitcoin script. (5)

Q.2(B) Explain Avalanche tree. (5)

Q.3 (A) What is an orphan block? How is it created? (5)

Q.3 (B) What is a ledger? Explain the concept of distributed ledger with its benefits. (5)

OR

Q.3 (A) Explain the concept of centralized ledger with its limitations. (5)

Q.3 (B) Explain Block Propagation. (5)

Section-II

- Q.4 (A) Explain consensus Mechanism in Blockchain. (5)
Q.4 (B) Explain Proof of work (POW) Mechanism. (5)
Q.4 (C) What is Mining? How Dose it work? (5)

OR

- Q.4 (C) Explain the concept of Mining pool. (5)
Q.5 (A) Explain Smart Contracts with respect to Blockchain Technology. (5)
Q.5 (B) Explain RAFT Algorithm with respect to permissioned Blockchain Technology. (5)

OR

- Q.5 (A) Explain Byzantine General Problem. (5)
Q.5 (B) How does Blockchain support Crowdfunding? (5)

- Q.6 (A) What is the Difference Between Bitcoin and Ethereum Blockchain? (5)
Q.6 (B) With respect to Ethereum Network, explain (a) Ethereum Virtual Machine (EVM), (5)
(b) Ether and Gas.

OR

- Q.6 (A) What is Metamask? What Is the Programming Language Used to Write Smart Contracts and Dapps? (5)
Q.6 (B) List at least 5 research domains/directions going on in the field of Blockchain. (5)

KADI SARVA VISHWAVIDHYALAYA
B.E. Semester VII Examination October-2023
CT703D-N Blockchain Technology

DATE: 30/10/2023

TIME: 12:00 pm to 03:00 pm

TOTAL MARKS: 70

Instructions:

- 1 Answer each section in separate answer sheet.
- 2 Use of scientific calculator is permitted.
- 3 All questions are **Compulsory**.
- 4 Indicate **clearly**, the options you attempt along with its respective question number.
- 5 Use the last page of main supplementary for **rough work**.

SECTION – I

- Q-1 A Compare Centralized, Decentralized and Distributed networks 5
B Explain the types of Blockchain. 5
C What is Genesis block. List and explain the components of a single block in Blockchain. 5

OR

- C What is an orphan block? 5

- Q-2 A Explain the terms Merkle root and Merkle tree. 5
B What are the characteristics of cryptographic hash function. 5

OR

- A Write short note on digital signature. 5
B Explain Public Key Cryptography. 5

- Q-3 A Explain Block propagation. 5
B What are the features of Bitcoin script. 5

OR

- A Write short note on Bitcoin. 5
B Explain distributed ledger in brief. 5

SECTION – II

- Q-4 A Write short note on mining. 5
B Explain the concept of distributed consensus in Blockchain. 5
C What is double spending? 5

OR

- C What is a mining pool? 5

- Q-5 A Explain proof of Work mechanism. 5
B Explain sybil and DDOS attack. 5
OR
A Write short note on Practical Byzantine Fault Tolerant Model. 5
B Write short note on proof of stake mechanism. 5
- Q-6A Explain the role of Blockchain in Medical health care. 5
B Write short note on smart contract. 5
OR
A Explain the role of Blockchain in compliance to KYC. 5
B What is the significance of Remix Platform in Blockchain? Also explain the term 'Gas' in smart contract execution. 5

KADI SARVA VISHWAVIDYALAYA
B. E. Semester – VII Examination (Nov-2022)
Blockchain Technology (CT703D-N)

Branch: CE/ IT/ CSE
Time: 10:00 am to 1:00 pm

Date: 11-11-2022
Total Marks: 70

Instructions:

1. Answer each section on separate answer sheet.
2. Use of scientific calculation is not required and hence not permitted.
3. All questions are compulsory.
4. Indicate clear, the option you attempt along with the respective question number.
5. All the acronyms are with respect to Blockchain and/or Networking.

Section – I

Q 1: (A)	List out various features of Blockchain Technology and explain <i>immutability</i> in detail.	5
Q 1: (B)	Explain the concept of centralization, decentralization and distributed.	5
Q 1: (C)	What are the different types of Blockchain?	5
OR		
Q 1: (C)	What are the Merkle trees? What is its importance in Blockchain?	5
Q 2: (A)	What is bitcoin? How did it evolve?	5
Q 2: (B)	Explain the double spending attack.	5
OR		
Q 2: (A)	How bitcoins are mines? How long does it take to mine a bitcoin?	5
Q 2: (B)	What are the features of bitcoin script.	5
Q 3: (A)	What is a ledger? Explain the concept of centralized ledger with its limitations.	5
Q 3: (B)	How chain is formed from Block? How transaction in any block can be detected?	5
OR		
Q 3: (A)	What is a ledger? Explain the concept of distributed ledger with its benefits.	5
Q 3: (B)	What is an orphan block? How is it created?	5

Section – II

Q 4: (A)	What is consensus in Blockchain? Why do we need it?	5
Q 4: (B)	List various consensus mechanisms of Blockchain and explain Proof of Work (PoW)	5
Q 4: (C)	What is Mining? How does it work?	5
OR		
Q 4: (C)	What is mining difficulty? How is it decided?	5
Q 5: (A)	Explain Byzantine General Problem.	5
Q 5: (B)	Explain Smart Contracts with respect to Blockchain Technology.	5
OR		
Q 5: (C)	Explain RAFT algorithm with respect to permissioned Blockchain Technology.	5
Q 5: (C)	Explain Crowdfunding with respect to Blockchain Technology.	5
Q 6: (A)	With respect to Ethereum Network, explain (a) Ethereum Virtual Machine (EVM), (b) Ether and Gas.	5
Q 6: (B)	What Is the Difference Between Bitcoin and Ethereum Blockchain?	5
OR		
Q 6: (A)	What Is Metamask? What Is the Programming Language Used to Write Smart Contracts and Dapps?	5
Q 6: (B)	List at least 5 research domains/directions going on in the field of Blockchain.	5

KADI SARVA VISHWAVIDHYALAYA
B.E. Semester VII Examination – October – 2021
CT703D-N Blockchain Technology

DATE: 29/10/2021

TIME: 10:00am to 1:00 pm

TOTAL MARKS: 70

Instructions:

- 1 Answer each section in separate answer sheet.
- 2 Use of scientific calculator is permitted.
- 3 All questions are **Compulsory**.
- 4 Indicate **clearly**, the options you attempt along with its respective question number.
- 5 Use the last page of main supplementary for **rough work**.

SECTION – I

- Q-1 A Compare centralized vs distributed network. 5
B Explain the types of Blockchain. 5
C Explain Genesis block and list the various components of a Block in a Blockchain. 5

OR

- C What is the difference between distributed Ledger and Blockchain? 5

- Q-2 A Explain the properties of cryptographic hash function. 5
B Explain Merkle tree. 5

OR

- A Explain Avalanche effect. 5
B List and explain the characteristics of Blockchain network. 5

- Q-3 A Explain consensus mechanism in blockchain. 5
B Explain proof of work (POW) mechanism. 5

OR

- A What is a mining pool? 5
B Explain proof of stake (POS) mechanism. 5

SECTION – II

- Q-4 A Write short note on mining in blockchain. 5
B Explain double spending problem. 5
C What is an orphan block? 5

OR

- C What is the significance of Remix Platform in Blockchain? Also explain the term 'Gas' in smart contract execution. 5

- Q-5 A Write short note on smart contract. 5
B Explain PBFT. 5
OR
A Explain proof of Burn mechanism. 5
B Explain sybil and DDOS attack. 5
Q-6A Write short note on Bitcoin. 5
B Explain the role of Blockchain in Medical health care. 5
OR
A Explain the role of Blockchain in compliance to KYC. 5
B Explain the permissioned blockchain consensus algorithms PAXOS and RAFT. 5

KADI SARVA VISHWAVIDHYALAYA
B.E. Semester VII Examination – March – 2023
CT703D-N Blockchain Technology

DATE: 28/03/2023

TIME: 10:00 am to 1:00 pm

TOTAL MARKS: 70

Instructions:

- 1 Answer each section in separate answer sheet.
- 2 Use of scientific calculator is permitted.
- 3 All questions are **Compulsory**.
- 4 Indicate **clearly**, the options you attempt along with its respective question number.
- 5 Use the last page of main supplementary for rough work.

SECTION – I

- Q-1 A Explain the types of Blockchain. 5
B Compare Centralized, Decentralized and Distributed systems. 5
C What are the characteristics of cryptographic hash function. 5
OR
C Explain Merkle tree. 5
- Q-2 A Write short note on digital signature. 5
B Explain double spending with an example. 5
OR
A Write short note on Bitcoin. 5
B What are the features of Bitcoin script. 5
- Q-3 A Explain Genesis block and list the various components of a Block in a Blockchain. 5
B Explain Avalanche effect. 5
OR
A Explain distributed ledger in brief. 5
B Explain Block propagation. 5

SECTION – II

- Q-4 A Explain the term consensus in blockchain. 5
B Write short note on mining. 5
C Explain the term orphan block. 5
OR
C Explain the concept of mining pool. 5

- Q-5 A Write short note on proof of work mechanism. 5
B Explain sybil and DDOS attack. 5
OR
A Write short note on proof of stake mechanism. 5
B Explain the benefits and challenges of a miner. 5
- Q-6A Write short note on PAXOS algorithm. 5
B Write short note on smart contract. 5
OR
A Explain proof of burn mechanism. 5
B Explain Byzantine General problem. 5